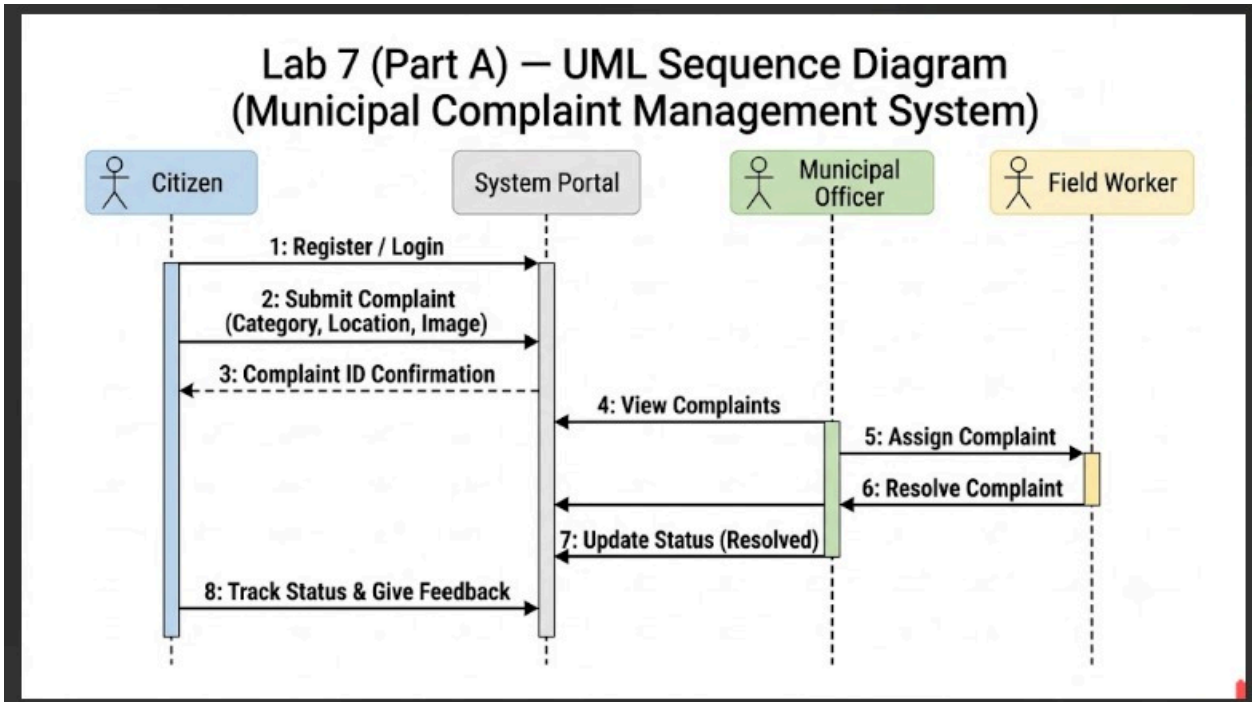
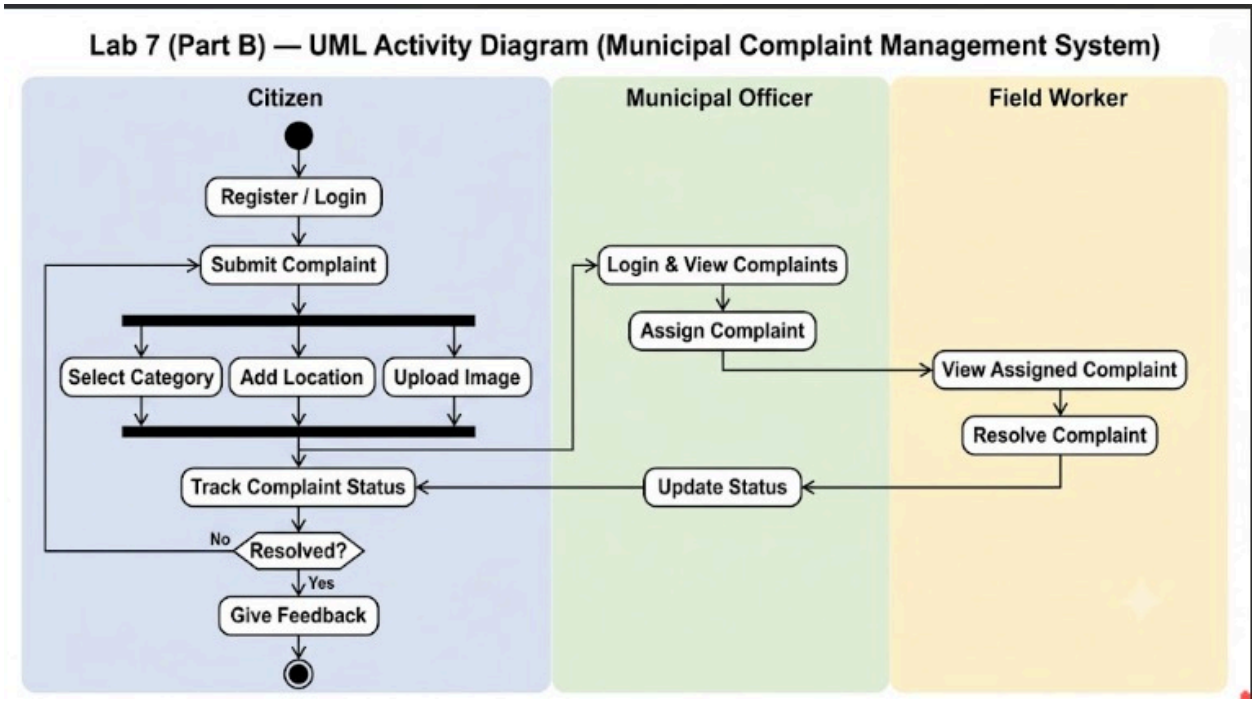




LAB 7 — UML SEQUENCE AND ACTIVITY DIAGRAM MODELING

3.1 Part A: UML Sequence Diagram Specifications (Dynamic Lifelines)

The sequence diagram details chronological message exchanges across four system lifelines: **Citizen**, **System Portal**, **Municipal Officer**, and **Field Worker**.

1. **Phase 1 (Submission):**
    - Citizen → System Portal: 1: Register / Login
    - Citizen → System Portal: 2: Submit Complaint (Category, Location, Image payloads)
    - System Portal → Citizen: 3: Complaint ID Confirmation
2. **Phase 2 (Administrative Assignment):**
    - Municipal Officer → System Portal: 4: View Complaints
    - Municipal Officer → Field Worker (via Portal): 5: Assign Complaint
3. **Phase 3 (Resolution & Tracking):**
    - Field Worker → System Portal: 6: Resolve Complaint (on-site closure)
    - Municipal Officer → System Portal: 7: Update Status (Resolved)
    - Citizen → System Portal: 8: Track Status & Give Feedback

3.2 Part B: UML Activity Diagram Specifications (Swimlane Workflow)

The activity diagram models parallel and sequential tasks partitioned across three distinct **Swimlanes**:

1. **Citizen Swimlane:**
    - **Initial Node** → Register / Login → Submit Complaint.
    - **Fork Node (Parallel Execution):** Concurrently runs Select Category, Add Location, and Upload Image, which merge back at a **Join Node**.
2. **Municipal Officer Swimlane:**
    - Control transfers to Login & View Complaints → Assign Complaint to Field Worker.
3. **Field Worker Swimlane:**
    - View Assigned Complaint → Resolve Complaint on Site.
4. **Validation & Closure Gateway:**
    - Officer executes Verify & Update Status.
    - Citizen monitors progress via Track Complaint Status.
    - **Decision Node ([Resolved?]):**
        - If *No*, loop routes back for re-checking or updates.
        - If *Yes*, citizen proceeds to Give Feedback, terminating at the **Final Node (End)**.